

An Entropy-based Analytic Model for the Privacy-Preserving in Open Data

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Abstract—In a Big Data era, a lot of open data set is published and shared with the public. That creates new services and business. However, the publication may cause a leakage problem of private information. In general, de-identification techniques are applied to the data before publication. The problem, however, has not been solved completely. Personal data can be obtained from the several sources such as Internet service and social media. In this situation, a de-identified open data may be simply joined with the leaked external data and it may result in a re-identification issue. We propose a new analytic model to measure the personal information leakage risk in the open data before publishing. The proposed model formulates the entropy-based re-identification risk to measure the privacy leakage risk. We also try to find the data utility measure by using the entropy while preserving the privacy. Based on both the risk and the utility measure, we propose the guideline for data open to the public. We show the guideline including the risk and utility measurement can be applicable with the empirical experiments.

Keywords—privacy; open data; re-identification; information entropy; privacy preserving data mining

I. INTRODUCTION

With the development of the Internet and wide use of mobile devices with sensors such as smartphone and IoT (Internet of Things) devices, various kinds of data are collected and the amount of the data becomes explosive. Storage, management and analysis systems for big data have been attractive and many developments are rushing to provide the variety of on/off-line services [1]. Big data has three characteristics (3Vs): volume, velocity, variety, and sometimes it is considered to have one more characteristic, value [2]. The advent of big data services has started opening data owned government, hospital and enterprise to the public. Governments collect and maintain personal, geographic and economic data. Governments promote ‘Open Government’ policy and boost the chances to create new business [3], through the public data portal. Hospitals and enterprises manage their customer information, and open the data for sharing the information with each other [4].

The data opening and sharing may have a risk of privacy leakage because opening the data can tempt malicious users to derive and guess the privacy data [5], [6]. The privacy data refers to the data that contains personal information or with which a person is identifiable. For an example, the name, the

cell phone number or the image of a person can be considered as a privacy data.

A typical solution to protect the private information prevents data identification, so called de-identifying the data before open to the public [7], [8], [9], [10]. The masking and reduction techniques are the representative solutions for de-identification. Though many de-identification techniques have been proposed, the big data environment makes it difficult to protect private information.

Nowadays, a lot of privacy data can be obtained from on-line through the Internet, social network and mobile services. Moreover, customers’ information governed by credit card firm may be leaked. In this environment, applying de-identification techniques to the “open” data is not sufficient to protect the privacy. Because the private information can be reproduced from the open data with the preexisting external data which includes the personal information. The process and techniques to identify the individual information in the de-identified data are called re-identification [6], [11], [12].

There are several well-known examples such as ‘Netflix’ case [13] and ‘AOL 4417749’ case [14]. Netflix Company revealed 500 thousands of de-identified rental records dataset. Unfortunately, it turned out to be possible to identify individuals by matching ratings between Netflix dataset and Internet Movie Database (IMDB). ‘AOL 4417749’ case happened when a research company AOL opened the text file which contains research history of 650 thousand users. In less than a week, the user of ID 4417749 was disclosed as 62 years old widow and her picture was published on the first page of New York Times. Likewise, even though it is difficult to deduce an individual only with de-identified data themselves, the other piece of external data makes it possible to find the privacy data, and the data can be abused for the unintended purposes.

In this paper, we propose an entropy-based guideline for open dataset. In the guideline, we introduce a re-identification risk model to prevent privacy data leakage from the open dataset. The model takes the information entropy as a measure of leakage risk of the private information in the open data. To better understand the re-identification risk of the whole dataset, our model figures out minimum entropy among all the entities in the dataset instead of the average entropy of them. We also propose a column-based re-identification risk model for more precise risk measurement and the guideline. An entropy-based

TABLE I. MEDICAL RECORDS OF A HOSPITAL

Name	Cell Phone Number	Zipcode	Gender	Age	Disease
Merry	480-292-5002	11611	F	28	Heart disease
James	520-220-2634	11611	M	31	Cancer
Terry	520-516-8600	11685	M	21	Diabetes
Eva	251-458-8924	11611	F	37	Cancer
John	251-623-2103	11701	M	50	Kidney disease
Mark	334-368-1952	11701	M	55	Heart disease
Jack	202-506-1602	11728	M	47	Diabetes
Jessica	202-341-5866	11728	F	49	Diabetes
Tom	202-709-0665	11685	M	29	Heart disease
Lily	208-592-2121	11611	F	23	Diabetes
Brad	208-573-7200	11685	M	36	Cancer
Julia	208-927-7662	11685	F	35	Cancer

TABLE II. DATA REDUCTION

Zipcode	Gender	Age	Disease
11611	F	28	Heart disease
11611	M	31	Cancer
11685	M	21	Diabetes
11611	F	37	Cancer
11701	M	50	Kidney disease
11701	M	55	Heart disease
11728	M	47	Diabetes
11728	F	49	Diabetes
11685	M	29	Heart disease
11611	F	23	Diabetes
11685	M	36	Cancer
11685	F	35	Cancer

data utility model is also proposed in order to promote the open data for public use or research while keeping privacy safe. Finally, we establish the guideline based on the risk and the data utility model. We experiment the guideline with the real dataset to show that our idea is applicable before publishing.

The rest of the paper is as follows. In Section II, we introduce the de-identification algorithms and the entropy. Section III explains our motivation and problem definition. We describe our analytic model in Section IV, and then we present our empirical experiment for our model in Section V. Section VI introduces previous work related to our model. Finally, we conclude this work in Section VII.

TABLE III. 4-ANONYMITY

Zipcode	Gender	Age	Disease
116**	*	2*	Heart disease
116**	*	2*	Diabetes
116**	*	2*	Heart disease
116**	*	2*	Diabetes
117**	*	>40	Kidney disease
117**	*	>40	Heart disease
117**	*	>40	Diabetes
117**	*	>40	Diabetes
116**	*	3*	Cancer
116**	*	3*	Cancer
116**	*	3*	Cancer
116**	*	3*	Cancer

TABLE IV. 3-DIVERSITY

Zipcode	Gender	Age	Disease
1161*	*	<40	Heart disease
1161*	*	<40	Cancer
1161*	*	<40	Cancer
1161*	*	<40	Diabetes
1170*	*	>40	Kidney disease
1170*	*	>40	Heart disease
1170*	*	>40	Diabetes
1170*	*	>40	Diabetes
1168*	*	<40	Diabetes
1168*	*	<40	Heart disease
1168*	*	<40	Cancer
1168*	*	<40	Cancer

II. BACKGROUND

A. De-identification Techniques

The publication of the data may cause the private information leakage. To prevent the leakage, the data should be published after all or part of the personal information is deleted or converted. These techniques are called de-identification.

Various de-identification techniques can be applied. Data reduction technique is a method for deleting the personal identification in the open dataset [15]. For example, there are hospital medical records as shown in Table I. In the records, “Identifier data” is defined as the information that can identify individuals such as ‘Name’, ‘Cell Phone Number’. “Sensitive

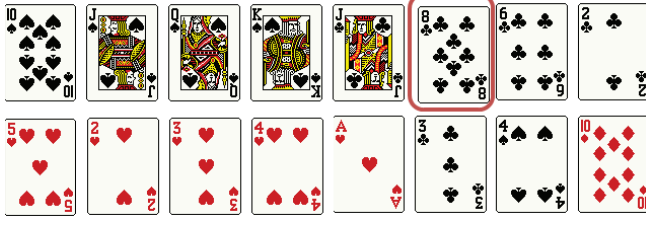


Figure 1 Example of a card game.

data” is also defined, such as ‘Disease’ that represents the critical personal information related to the identifiable data. By applying the data reduction technique to Table I, the identifier data (Name and Cell Phone Number) is deleted, and only the sensitive data (Disease) and the other information are published (Table II). The reduction technique can hide the information that someone is taken with a certain disease. However, the reduction technique is not good enough to reduce the privacy leakage risk.

In Table II, we can easily identify the identifier data from the combination of three attributes (Zip code, Gender, Age). The combination is called the quasi-identifier [5]. The quasi-identifier should be considered. The data masking technique is proposed to anonymize them. The data masking divides the data into groups by hiding a portion of the quasi-identifier. The well-known masking algorithms are k -anonymity [7] and l -diversity [8]. In the k -anonymity algorithm, the quasi-identifiers are masked in order that the number of data in a group should be at least $k-1$. Table III shows the result of the k -anonymity technique ($k=4$) applied to the quasi-identifiers in Table II, and four data are included at least in a group. However, the k -anonymity technique does not completely avoid identifying personal information. In Table III, the last group has the only type of the sensitive data ‘Cancer’. In such a situation, we can easily find the people taken with cancer who are the people in their 30s and live in the 116** area. The problem can be coped with the l -diversity technique. The technique enforces each group to be composed of at least l different sensitive data. After applying it to Table II, the sensitive data in Table IV, are divided into three groups and each group has at least three different sensitive data, that is, ‘Disease’ value ($l=3$).

B. Entropy

1) *Information Entropy*: Information entropy is a measure of the information uncertainty used in information theory. Since Claude E. Shannon has introduced the entropy which is called Shannon entropy in 1948 [16], it is widely accepted. The entropy represents the uncertainty of the information corresponding to an event occurred in a certain probability. It means that the entropy is how much information we need to identify data. In other words, the amount of information corresponds to the uncertainty for the occurrence of some events. The higher entropy produces the higher randomness of events, while the lower entropy produces the higher predictability of events.

For example, if we play the game to find a selected card of the 16 cards in Figure 1, we can guess the selected card through the following question process.

- 1) “Is the card in the right half? Yes.”
- 2) “Is it in the top of the right half? Yes.”
- 3) “Is it in the right half of the top? No.”
- 4) “Is it to the right of the left half of the top? Yes.”

Thus, we can answer that the selected card is ‘Clover 8’ through the 4 questions, and the value of 4 is the information entropy of this game. The equation for the entropy is written as:

$$H(X) = -\sum_{x \in X} p(x) \log_2 p(x) \quad (1)$$

The probability $p(x)$ is that the chance of x is in the probability distribution X , and $\sum_{x \in X} p(x) = 1$. In the above example in Figure 1, if there are 16 cards, the probability of selecting a card is $1/16$. And thus the entropy for this card game is 4 by (1).

2) *Kullback-Leibler Divergence*: In Information Theory and Probability Theory, the Kullback-Leibler Divergence is a function to calculate the difference in the entropies between two probability distributions [17]. The divergence is called the relative entropy, information gain or information divergence. If $p(x)$ and $q(x)$ are the probabilities of the probability distributions P and Q respectively, Kullback-Leibler Divergence of P , Q is defined as shown below.

$$\begin{aligned} D_{KL}(P||Q) &= \sum_x p(x) \log_2 \frac{p(x)}{q(x)} \\ &= (-\sum_x p(x) \log_2 q(x)) - (-\sum_x p(x) \log_2 p(x)) \\ &= (-\sum_x p(x) \log_2 q(x)) - H(X) \end{aligned} \quad (2)$$

Since Kullback-Leibler Divergence that is not a distance of P, Q has the asymmetric value, the result depends on the position of two values P, Q . The following is the characteristics of Kullback-Leibler Divergence.

- $D_{KL}(P||Q) \geq 0$
- $D_{KL}(P||Q) = 0$ iff $p(x) = q(x)$ for all $x \in X$
- $D_{KL}(P||Q) \neq D_{KL}(Q||P)$

III. MOTIVATION

When the data are published, the de-identification techniques should be adopted to prevent the leakage of privacy. However, the data including the personal information through Internet service and social media is revealed, and the leaked customer data of credit card firm may be caught. The data may include someone’s name, birth date, cell phone number, age and credit card number. The big data environment includes both “open” data and external data (Figure 2).

In the environment, the open data still has the risk of private information leakage even if the personal information is de-identified and published because the de-identified data can be identified by joining with the external data. This process or technique is called re-identification [11]. In other words, the problem of revealing the sensitive data is raised by

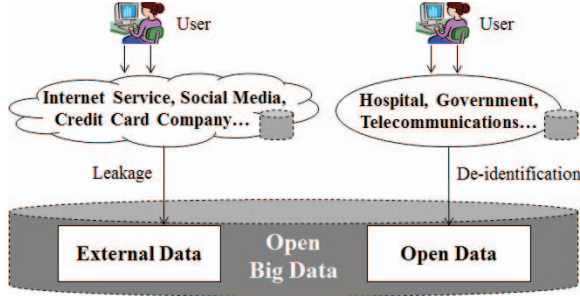


Figure 2 Open big data environments.

the re-identification technique that rebuilds the relationship between the external data and the sensitive data. In order to solve this problem, we need to know the private information leakage of the de-identified data before opening the data. Namely, it is necessary to quantify the re-identification risk of the open data connected with the external data.

The following problems should be addressed to cope with the above situation. First, we need to know how to identify the re-identification risk and how to quantify the re-identification risk. Second, we need to define the form of the external data which includes the personal information. There are almost unlimited external data. It is practically impossible to determine the re-identification risk with all of the available external data in the world. We should simply redefine the external data, and use it to measure the re-identification risk.

IV. ENTROPY-BASED ANALYTIC MODEL

A. Preliminary

Re-identification is a collection of techniques by matching the external data with the de-identified open data. Namely, it is possible to reveal the sensitive data in the open data joining them with the identifier data of the external data. Now, an entity of the open data consists of the sensitive data attribute and the other attributes, which is defined as *S-entity*. *S-attribute-set* is defined as the subset of attributes which includes the sensitive data attribute. The entity of the external data is defined as *I-entity*. *I-entity* consists of the attributes in *S-attribute-set* excluding the sensitive data and the attributes whose value is the identifier data. Figure 3(a) contains the hospital medical records whose disease value is the sensitive data. An entity containing all the attributes is an *S-entity*, while an *S-attribute-set* consists of the attributes disease and age 30s. The Figure 3(b) shows the external data. The entity of ‘Terry’ which has an identifier data attribute (Name) and other attributes is an example of *I-entity*.

B. *I-entity* Dataset

The set of *I-entities* (external data) reveals the personal information on/off-line. The personal information usually contains the name, the cell phone number of individuals, which easily identifies them. The *I-entity* dataset plays a role in joining the sensitive data in the *S-entity* with the identifier data in *I-entity*, which provokes re-identification issue. Both the *I-entity* and the sensitive data in *S-entity* are needed to measure re-identification risk more precisely. However, it is

Name	Cell Phone Number	Zipcode	Gender	Age
Merry	480-292-5002	11611	F	28
James	520-220-2634	11611	M	31
Terry	520-516-8600	11685	M	21
Eva	251-458-8924	11611	F	37
John	251-623-2103	11701	M	50
Mark	334-368-1952	11701	M	55
Jack	202-506-1602	11728	M	47
Jessica	202-341-5866	11728	F	49
Tom	202-709-0665	11685	M	29
Lily	208-592-2121	11611	F	23
Brad	208-573-7200	11685	M	36
Julia	208-927-7662	11685	F	35

hard to figure out the *I-entity* dataset potential exterior. We face how to get the *I-entity* dataset for measuring the re-identification risk.

The *I-entity* data have different kinds of types and comes from multiple sources. The *I-entity* data can be joined with the other *I-entity* data. In the worst case, joining among them can produce the data almost same as original ones before de-identification process [18]. In this paper, the *I-entity* data can be considered as the data excluding the sensitive data from the original data. For example, the de-identification process with the original data in Table I results in the data in Table II. The *I-entity* dataset relevant with Table II contains all the data except the disease data, which is the sensitive data (Table V).

C. Entropy-based Re-identification Risk

We propose an entropy-based re-identification risk measuring model for the “open” data (*S-entity* dataset). The measurement method adopts the information entropy and evaluates the re-identification risk of each *I-entity* with *S-entity* dataset. Based on the result, we can figure out the overall re-identification risk.

We start with calculating the amount of matched information between the identifier data in *I-entity* and the sensitive data in *S-entity* by applying information entropy measurement. The amount of matched information is defined in (3).

$$H(G|i) = -\sum_{s \in G} p(s|i) \log_2 p(s|i) \quad (3)$$

G represents the set of *S-entities* and s is an *S-entity* in G . Also, K represents the set of *I-entities*, where i means each *I-entity* in K . $p(s|i)$ means the probability of matching the identifier data of *I-entity* i and the sensitive data of *S-entity* s . The equation refers to the amount of information to predict the relevance of the identifier data to the sensitive data. In other words, $H(G|i)$ is the matching entropy to identify that the identifier data of i is connected with which sensitive data in G .

TABLE V. THE EXAMPLE OF *I-ENTITY* DATASET (EXTERNAL DATA)

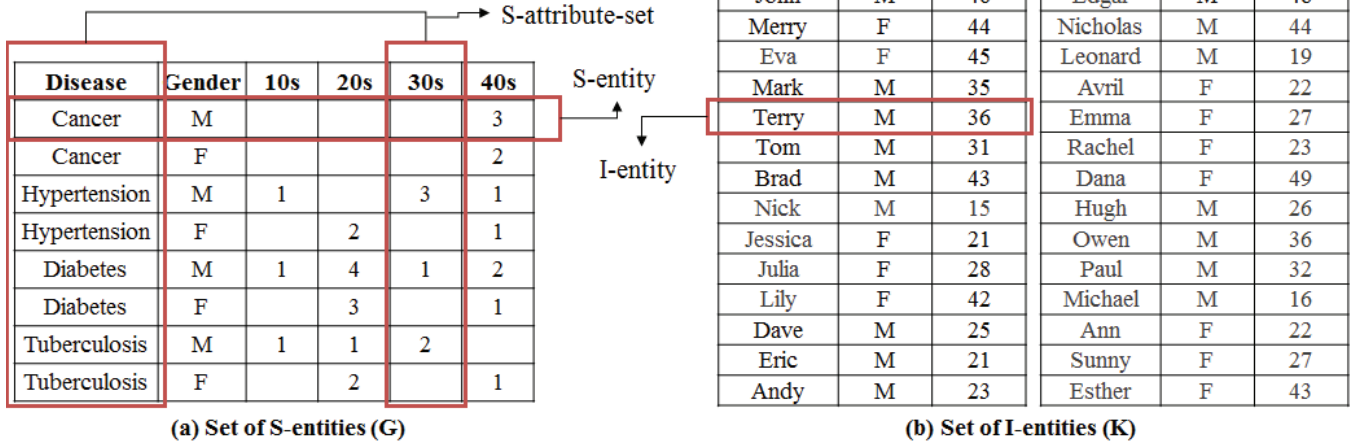


Figure 3 The example of *S-entity* dataset (G) and *I-entity* dataset (K).

We explain the equation with Figure 3. There is the *S-entity* dataset (G) and the *I-entity* dataset (K). G is the published from the medical records of a hospital while K appears to be original medical data excluding the disease values which are the sensitive data. Then we can calculate the matching entropy of 'James' *I-entity*, $H(G|i = \text{James})$.

$$H(G|i = \text{James}) = -\frac{3}{6}\log_2 \frac{3}{6} - \frac{1}{6}\log_2 \frac{1}{6} - \frac{2}{6}\log_2 \frac{2}{6} = 1.4591$$

The probability that 42 years old 'James' gets cancer is 3/6, hypertension 1/6, diabetes 2/6 and tuberculosis 0. $H(G|i = \text{James})$ is 1.4591.

We also measure the maximum entropy which represents the maximum uncertainty of the information corresponding to an event occurred in a certain probability. Equation (4) shows how to calculate maximum entropy.

$$H_{\max}(X) = -\sum_{x \in X} \frac{1}{n} \log_2 \frac{1}{n} \quad (4)$$

The maximum entropy is considered as the maximum amount of required information to determine whether the sensitive data is relevant to the identifier data in *I-entities*. In Figure 3, the number of the sensitive data values is four: cancer, hypertension, diabetes, tuberculosis. $H_{\max}$, the maximum entropy of the relation between the identifier data in *I-entity* dataset and the sensitive data in *S-entity* dataset, is $-\sum_{i=1}^4 \frac{1}{4} \log_2 \frac{1}{4} = 2$.

To apply the matching entropy to our re-identification measuring model, we adopt the relative entropy called Kullback-Leibler Divergence technique [17]. The relative entropy is measured between the maximum entropy and the matching entropy. $D_{KL}(i||\max)$ shown in (5) indicates how much difference the matching entropy of *I-entity* i from the maximum entropy. It represents the dependency of the re-

identification risk on the relative entropy between the maximum entropy and the matching entropy.

$$D_{KL}(i||\max) = \sum_{s \in G} p(s|i) \log_2 \frac{p(s|i)}{q(\max)} = (-\sum_{s \in G} p(s|i) \log_2 q(\max)) - H(G|i) \quad (5)$$

The re-identification risk (R_i) of *I-entity* i with G can be obtained from the maximum entropy and the relative entropy. Because the re-identification risk increases as much as the amount of removed information, the re-identification risk of *I-entity* i , R_i can be defined by (6).

$$R_i = \frac{D_{KL}(i||\max)}{H_{\max}} \times 100 \quad (6)$$

$D_{KL}(i||\max)$ and R_i of 'James' in Figure 3 are calculated as 0.5409 and 27.04% respectively.

$$D_{KL}(i = \text{James}||\max) = \left(-\frac{3}{6}\log_2 \frac{1}{4} - \frac{1}{6}\log_2 \frac{1}{4} - \frac{2}{6}\log_2 \frac{1}{4}\right) - 1.4591 = 0.5409$$

$$R_i = \frac{D_{KL}(i = \text{James}||\max)}{H_{\max}} \times 100 = \frac{0.5409}{2} \times 100 = 27.04\%$$

After calculating the re-identification risk of each *I-entity*, previous algorithms mostly obtain the average value of them for the risk of the whole dataset. However, the average value is difficult to determine the accurate risk of *I-entity* dataset because the risk value of each *I-entity* smoothly melts into the

Disease	Gender	10s~20s	30s~40s
Cancer	M	0	3
Cancer	F	0	2
Hypertension	M	1	4
Hypertension	F	2	1
Diabetes	M	5	3
Diabetes	F	3	1
Tuberculosis	M	2	2
Tuberculosis	F	2	1

Figure 4 The merged data from Figure 3(a).

average value. To solve this problem, the maximum value among re-identification risks of all the *I*-entities can be considered as the re-identification risk of the whole dataset. In other words, when *G* is published to the public, the highest risk of re-identification among *I*-entities should be chosen in *K*.

$$R(G|K) = \max_{i \in K} R_i \quad (7)$$

Equation (7) means that the *G* contains an *I*-entity whose maximum re-identification risk is $R(G|K)$. Even though average re-identification risk is low, the weakest point corresponding to the maximum risk can avoid re-identification risk at maximum.

In Figure 3, by calculating the average, the re-identification risk of *G* is 27.52

%. It means that the hospital medical dataset *G* has a low risk of privacy leakage. However, the re-identification risk of ‘Dave’ entity, which contains the data of a man in his 20s, is 63.90%. Consequentially, we could not detect the riskiness of dataset *G* through average value. By evaluating maximum risk, the re-identification risk of *G* in Figure 3 is 63.90% because the re-identification risk of the man in his 20s entity is maximum. The distribution of the men in their 20s is the most unbalanced.

D. *S*-attribute-set based Re-identification Risk

We propose another model to figure out which attribute set is weak on re-identification issue. The model is called ‘*S*-attribute-set based re-identification risk’. In Figure 3, an *S*-attribute-set consists of the sensitive data attribute (Disease) and the other attributes: ‘Disease-10s’, ‘Disease-20s’, and ‘Disease-30s’. The re-identification risk of the *S*-attribute-set can be measured by (7). With the column-based technique, we can calculate the entropy for smaller units of the dataset than the row-based. In other words, it means that the re-identification risk of the whole dataset is more precisely determined.

From the re-identification risk of each *S*-attribute-set, we can identify the riskiest attribute in the *S*-entity dataset. After the riskiest attribute is deleted or included in the other

attribute, the entropy of *G* becomes higher and the re-identification

Name	Gender	Age	Disease
James	M	42	Cancer
Jack	M	45	Cancer
Andy	M	23	Diabetes
Merry	F	44	Cancer
Eva	F	45	Cancer
Mark	M	35	Hypertension
Alex	M	29	Diabetes
Julia	F	28	Hypertension
John	M	46	Cancer
Nick	M	15	Hypertension
Jessica	F	21	Hypertension
Tom	M	31	Hypertension
Lily	F	42	Hypertension
Dave	M	25	Diabetes
Eric	M	21	Diabetes
...	...	...	...

Figure 5 The portion of the original data corresponding to Figure 3(a).

harder in the de-identified *S*-entity dataset. Hence, the re-identification risk measurement of each the *S*-attribute-set makes it possible to identify which attributes are vulnerable. It means that our model provides the guideline to reduce the risk by de-identifying the vulnerable attributes.

From Figure 3, the probability of ‘Disease-20s’ *S*-attribute-set gets cancer is 0, hypertension 2/12, diabetes 7/12, and tuberculosis 3/12. The re-identification risk is 30.77%. We can also measure the re-identification risk of the other *S*-attribute-sets: ‘Disease-10s’ is 20.75%, ‘Disease-30s’ is 27.04%, and ‘Disease-40s’ is 10.5%. The ‘Disease-20s’ and ‘Disease-30s’ *S*-attribute-sets show the highest risk among all the *S*-attribute-sets, and the two *S*-attribute-sets would be merged with the other *S*-attribute-sets such as ‘Disease-10s’ and ‘Disease-40s’, respectively (Figure 4). As the result, the re-identification risk decreases from 63.90% to 35.06%.

E. Entropy-based Data Utility

If the de-identification process of an *S*-entity dataset has been done, the privacy leakage risk can decrease. But it may become hard to obtain the useful information from the *S*-entity dataset. The privacy and the data utility are in trade-off relationship [19]. We have to optimize the relationship between the privacy and the utility. Thus, we need a data utility measurement model to quantify the data utility.

The data utility can be considered as the difficulty of obtaining the accurate query results from the dataset. It means how much information (entropy) we need to obtain the query results in the dataset. The higher entropy produces the lower data utility, while the lower entropy produces the higher data utility. From this idea, we propose an entropy-based data utility model for the *S*-entity dataset. The model measures the data utility by ratio the difference between the entropy of the

S -entity dataset and the maximum entropy to the maximum entropy as (9) shows. The S -entity dataset's entropy, which is

Disease	Gender	00-14	15-18	19-29	30-39	40-49	50-59	60-64	65-
Alcoholism	M	0	0	0	295	142	132	0	103
	F	0	0	0	0	0	3	0	0
Dysthymia	M	0	45	0	0	0	0	0	0
	F	78	0	0	0	0	0	0	0

⋮
(a) D_1

Disease	Gender	00-18	19-29	30-39	40-49	50-59	60-64	65-
Alcoholism	M	0	0	295	142	132	0	103
	F	0	0	0	0	3	0	0
Dysthymia	M	45	0	0	0	0	0	0
	F	78	0	0	0	0	0	0

⋮
(b) D_2

Disease	Gender	00-18	19-29	30-39	40-49	50-59	60-
Alcoholism	M	0	0	295	142	132	103
	F	0	0	0	0	3	0
Dysthymia	M	45	0	0	0	0	0
	F	78	0	0	0	0	0

⋮
(c) D_3

Disease	Gender	00-18	19-29	30-49	50-59	60-
Alcoholism	M	0	0	437	132	103
	F	0	0	0	3	0
Dysthymia	M	45	0	0	0	0
	F	78	0	0	0	0

⋮
(d) D_4

Disease	All age
Alcoholism	675
Dysthymia	123

⋮
(f) D_6

Disease	Gender	All age
Alcoholism	M	672
	F	3
Dysthymia	M	45
	F	78

⋮
(e) D_5

Figure 6 The partial results of the de-identification methods applied to the real data.

defined by (8), for the data utility is an average value of all the I -entities.

$$H(G) = \text{avg}\{H(G|i)\} \quad (8)$$

$$U = \frac{H_{\max} - H(G)}{H_{\max}} \times 100 \quad (9)$$

In the original data shown in Figure 5, the result of a query 'the number of men in 25 ~ 32years old who suffer from diabetes' is 2, 'Alex' and 'Dave'. The entropy of the original data is 0 because we can obtain the accurate results directly without any information. But, the result of the query is 5 in the S -entity dataset in Figure 3. It means that the accurate results of the S -entity dataset are able to be obtained with the additional entropy 0.9453. Consequentially, the data utility of the S -entity dataset is 52.73%.

V. EMPIRICAL STUDY

To show the measurement models of the re-identification risk, and the data utility are effective, we experimented the model with real medical data. We obtain an S -entity dataset, which is partially represented in Figure 6(a), from Korean 'Open Data Portal' [20]. We also build the I -entity dataset, which is about 60,000 medical records with name, age, and gender but without the disease, from the S -entity dataset. The attribute name is filled with random values. The attribute age

has the value from 0 to 80. There are 61 diseases such as Alcoholism, Dysthymia, etc.

We apply six de-identification techniques ($D_1 \sim D_6$). Figure 6 shows the partial results of six different de-identification methods applied to the experimental data. D_1 , which is the real data obtained from the portal, is the result from grouping data by disease, gender and age, and counting the number of patients for each group. D_2 merges the two attribute groups of D_1 , '00-14' and '15-18' because the risk of the S -attribute-set 'Disease-(00-14)' is the highest value in D_1 . By our S -attribute-set based re-identification risk model, D_3 and D_4 also merge the attributes '60-64' $\cup$ '65-' and '30-39' $\cup$ '40-49', respectively. D_5 merges all the 'age' attributes based on D_4 . D_6 has only the attribute 'disease' and the values are aggregated. D_1 is applied to de-identification at least and D_6 at most.

Figure 7 shows both the privacy and the data utility measures of the six different de-identification. The privacy value is obtained from subtracting the re-identification risk (%) from 100. The result indicates that the more data is de-identified, the higher privacy can be achieved. In contrast, the more data is aggregated, the less data utility is provided. In our experiment, the published real dataset D_1 has the low privacy even though it is very useful. For this reason, our model sequentially suggests the de-identification methods from D_2 to D_6 for the open dataset. Among these methods, the data utility is less than the privacy from D_5 . Accordingly, we

recommend choosing the D_4 policy to balance the relationship between the privacy and the data usability. The privacy of D_4

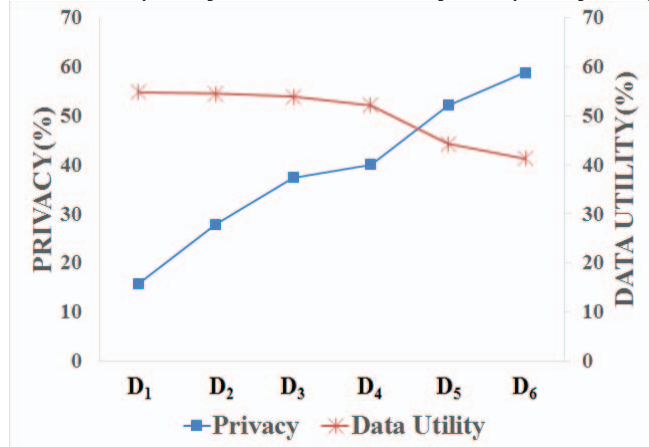


Figure 7 Privacy and data utility measure of the various de-identifications.

is increased by about 35 percent compared to D_1 while the data utility is decreased by about 2 percent. In other words, when the D_1 dataset published, the proposed guideline can advise that the de-identification technique from D_4 has lower re-identification risk than D_1 with slightly diminishing the data utility. On the contrary to this, if the higher data privacy is required, then D_5 might be the better option to reduce re-identification risk.

VI. RELATED WORK

Information entropy can easily refer to the amount of information numerically and identify a change in the amount. To protect the privacy, a lot of researches use the entropy to determine re-identification risk [18], [21], [22].

In [18], the information entropy is used to measure the value of k which makes the lowest re-identification risk of the de-identified data by the k -anonymity method. Michele also defines the leaked data for measuring the re-identification risk. Mousa et al. [21] take the entropy to propose the disclosure risk model. The model measures the risk of the data which is de-identified by query restriction, sampling, and noise addition. [22] identifies the re-identification risk of the location-based data from various sources. The re-identification risk is determined by measuring the amount of location information based on the entropy. Because the previous studies use the information entropy, they can calculate the re-identification risk of the data which is applied to any de-identification techniques.

However, the previous studies have not addressed the following problems. First, we cannot identify the re-identification risk of the internal entity whose personal data can be easily leaked. The previous studies evaluate the total entropy of the entire entities. Because the entropy of an entity is merged into the total entropy, it is hard to identify the riskiest entity. Second, the previous studies concentrate more on measuring the re-identification risk than the data utility. The data utility is also required to promote the open data for public use while keeping privacy safe.

In comparison, we propose both the entropy-based data utility model and re-identification model to determine all the entropy values of all the entities.

VII. CONCLUSION AND FUTURE WORK

These days, there is a huge amount of data open to the public, but the open data has the potential risk of the re-identification issue with the external data which includes the personal information. To cope with the situation, we propose an entropy-based risk model to measure the degree of the re-identification risk in advance before publishing the data. The model takes S -attribute-set as a base unit to measure the entropy of an entity, and the entropy of the data can be obtained from all the entity's risk. We can achieve the more accurate re-identification risk model based on the fine unit. Also, by suggesting an entropy-based data utility model for the open data, not only the privacy is protected, but also the data usability would be guaranteed. In our experiments, our model recommends an effective de-identification technique whose re-identification risk is decreased by about 35 percent and data utility is decreased by only about 2 percent.

The proposed models draw two values, the re-identification risk and the data utility separately. These two values have the trade-off relationship. We are going to develop an integrated model including both the privacy and the data utility to recommend an optimum de-identification policy.

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